

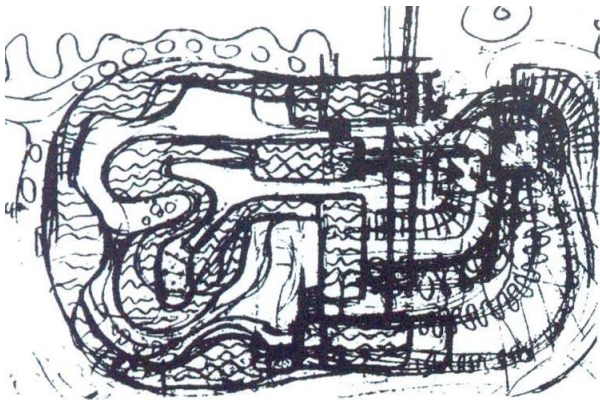
Warmonderhof 1976.

Due to interest in such systems and their application in different fields, a student at Emerson College, Nico Torenstra, called in Prof Mansvelt who was Director of the State supported Warmonderhof Biodynamic Farm training Institute near Tiel in Holland.

It was decided to set up a research installation connected to effluent discharge on the Institute campus. My initial sketch plan was followed, providing two separate biological systems to study the contrast between a Step Cascade and Flowform Cascade treatments. The former movements being dominated by gravity, the latter by rhythms and lemniscatory processes, which tended to alternately allow water to fall and rise within the Flowforms.

Over a period of some four years it was established that a main tendency on the influence upon plant growth from the Step Cascade was a vegetative enhancement and due to the Flowform treatment a generative enhancement. This means that the latter plants bore increased blossom and seed production. Today the flourishing installation is used by a housing association to deal with their wastewater.

original plan



inaugurated 1976



Shortly after construction



As it is functioning today after 33 years

Summary of findings from Warmonderhof University project

Findings:

Flowform cascade produced earlier and stronger flowering with plant morphology as if more light available, eliminated smell of ammonia with successful biological waste water purification resulting in an upstream waterfall type environment.

Detailed results:

Measurement:	Stepped Cascade	Flowform Cascade
Plant development:	vegetative growth corresponding to eutrophic (downstream, delta) of a river system wide leaf production normally found in shaded sites	generative growth corresponding to oligotrophic (upstream, waterfall) of a river system flowering earlier in the summer, colouring more deeply in autumn, stressing upright branches and smaller leaf production, normally found in lighter sites
Macrofauna composition:	dark habitat, slower movement, life cycle that includes flying phase	lighter habitat, faster and more nervous movement, lifecycle that remains in water – no mosquitos
Direct observations:	higher turbidity, smell of ammonia	lower turbidity (clearer water), smell of hay & hummus
Drop picture evaluation:	less pronounced structure – lower quality	more pronounced structure – higher quality
Incidental observations:	slow and passive movement of fish	more playful and active





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TO WHOM IT MAY CONCERN,

SUMMARY: In a research project of four years at Warmonderhof, "polishing" influences of Virbela Flow Form water movement on biological water quality could be demonstrated and tentatively characterized. The so-called Malmö type of Flow Form was taken as a representative of the Virbela Flow Forms.

RESEARCH AIM: Research was done to find out if any effects of a Flow Form treatment on organically polluted waste water could be demonstrated and - if so - how such effects could be evaluated in terms of water quality.

RESEARCH METHODS: Virbela Flow Form Cascade treated water was compared with Stepped Cascade treated water in parallel systems that differed only in the ways in which the water moved over and through those cascades:

- strongly turbulent in rhythmically alternating vortices, versus:
- smoothly and steadily floating down in a more laminary way.

The width of the steps was about the same size as that of the widest part of the tested Malmö Flow Forms.

Chemical parameters usual in Water Pollution Control were combined with biological parameters focussing on plant development and macrofauna com-

position, completed by direct observations on water colour and smell and also by so-called drop-picture-analysis according to Schwenk.

Regularly repeated observations were made in the two parallel series of three ponds, each fed with treated water from one cascade and furnished each spring with similar groups of selected water/bank plants from natural sites in early stages of their annual development. Macrofauna developed "spontaneously" out of the treated water, the rootsystems of the imported plants and from the air (wind, birds, etc.).

RESULTS: In the four yearly repetitions a general trend in developmental differences between the compared systems could be established, which can be stated as follows:

In the Stepped Cascade System:

- plant development biased towards vegetative growth, stressing wide leaf production, somewhat in the way of plants growing in shaded sites;
- macrofauna composition biased towards species that prefer a darker habitat (deep water and bottom), display a somewhat softer-rounded exterior shape, slower movements and a life cycle that includes a flying phase [f.e. midge larvae];
- direct observations showed a tendency to higher turbidity and a smell more like fug-must-ammonia;
- drop pictures displayed a less pronounced structure.

In the Virbela Flow Form System:

- plant development biased towards generative growth, flowering earlier in summer, colouring more deeply in autumn, stressing upright branches and smaller leaf production, somewhat in the way of plants growing in lighter sites. However, here as well as in the Stepped Cascade ponds, the whole water surface was covered with plants of the mentioned phenotypes.
- macrofauna composition biased towards species that prefer a lighter habitat (upper water layers and surface), display a more pronounced and rugged exterior, faster and more nervous movements and a life cycle that remains in the water [f.e. crustaceans, watermites];

- direct observations showed a tendency to lower turbidity (clearer water) and a smell more like humus, eventually hay;
- drop pictures displayed more pronounced structure.

Incidental observations on goldfish behaviour showed them to be more playfully active near the turbulent outlet of the Virbela Flow Form Cascade and more slow and passive near the tardy outstream of the Stepped Cascade.

Over the four years of chemical analysis of the water of both systems, no consistent differences were found. Their temporarily fluctuations could not yet be correlated to the described trends in ecosystem development. When length of streamline and height of the cascade were similar, oxygen-up-take in similar amounts of water was alike in both multi-step systems. A tendency towards higher oxygen efficiency in the Flow Form System was noted but could not be proved in a satisfying way.

EVALUATION: Both systems proved to be usefull for the oxygen-uptake necessary for biological waste-water purification.

However: purifying processes (breakdown of organic matter, etc.) were directed in different ways under the influences of the different water movements.

The more tardy and laminary water movements in the Stepped Cascade directed those purification processes in such a way that a vegetative growth was stimulated in the ecosystem of the pond, comparable to the one in an overshadowed, calm and wide or downstream, eutrophic part of a river system.

The livelier, rhythmically pulsating and turbulent water movements in the Flow Form System stimulated the development in the ecosystem of the pond toward generative growth, comparable to that in a light, open rapid - or waterfall - in an upstream, oligotrophic part of a river system.

When drinking-water - meant for human consumption - should be appropriate for stimulation of the development of higher living organisms, Virbela Flow Forms could be worth to be incorporated in the purification system.

Where turbulent (extravert) phases and tardy (intravert) phases alternate in biological water purification systems, the light-open turbulence of the Virbela movements might be an important extra polishing treatment, re-integrating the water into the natural environment from which it was technologically isolated for human use.

However, as to the special place(s) in the systems where those Flow Forms would fit best, full scale trials should be made in practice.

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- please see figures following behind -

Figure 1.

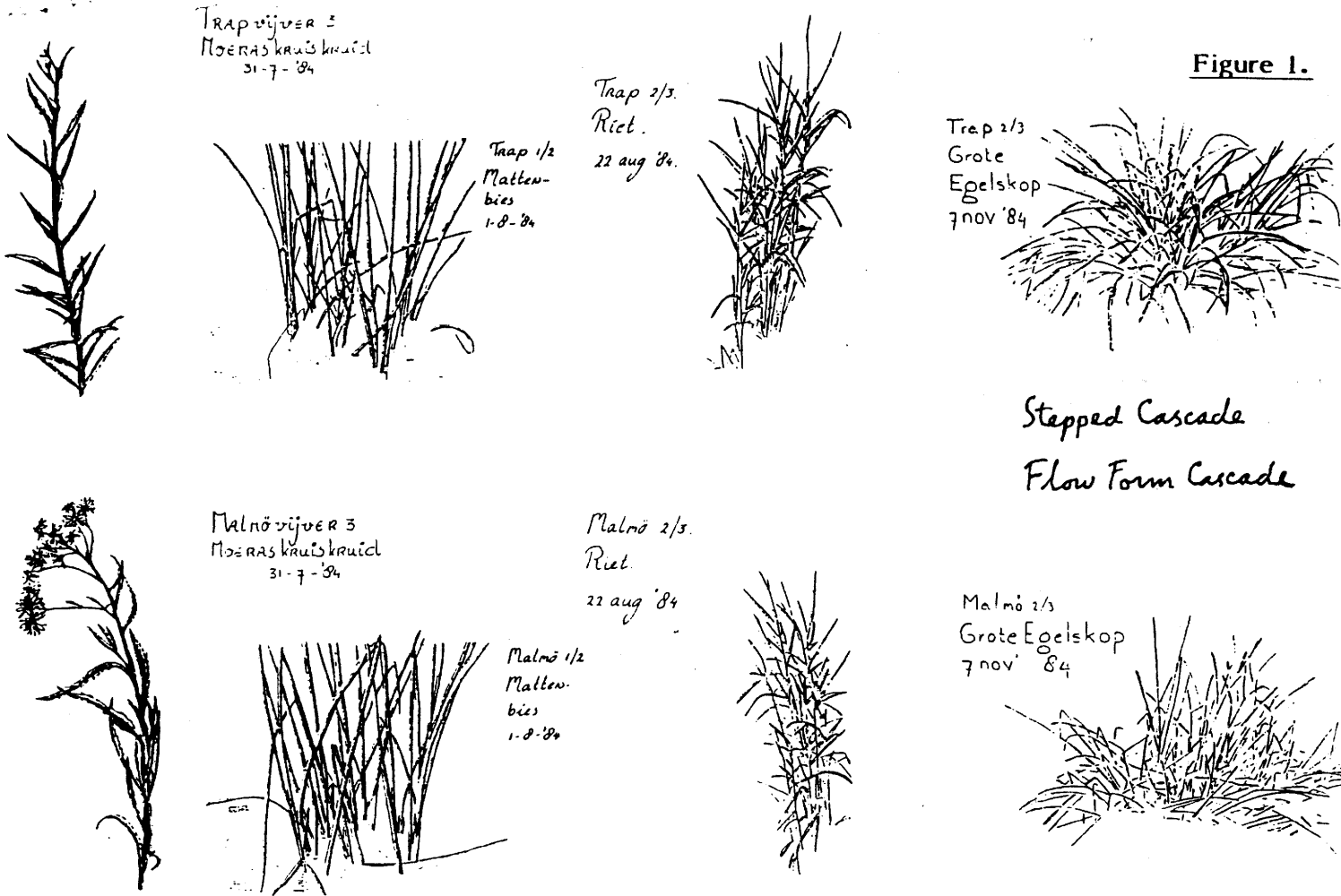


Figure 2.

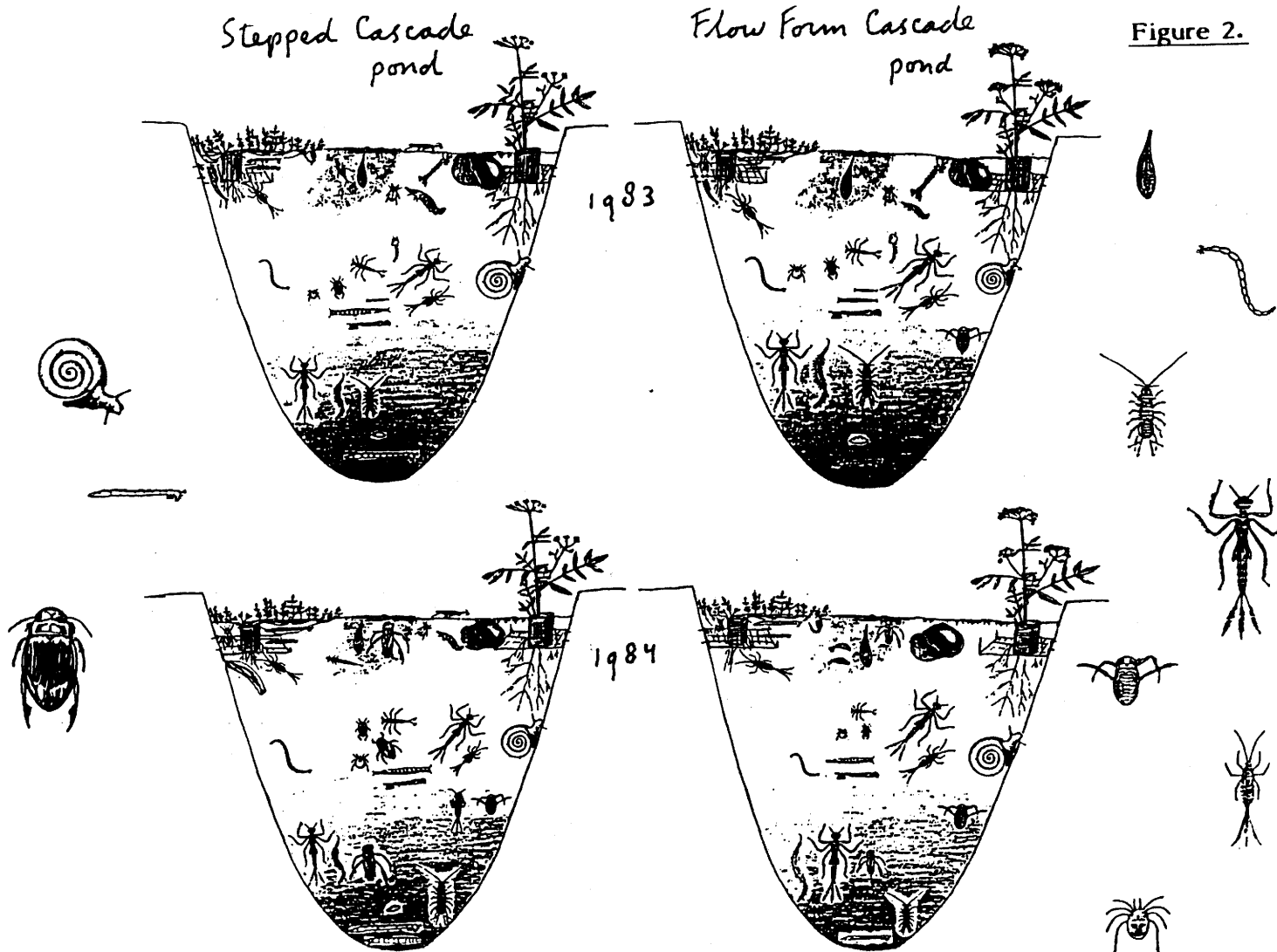


Figure 1. Some examples of characteristic phenotype-differences between plants grown in the Stepped Cascade ponds (upper row) and those grown in Flow Form Cascade ponds (lower one). From left to right: Senecio paludosus, Scirpus lacustris, Phragmites australis, Sparganium erectum. (See also Figure 3.)

Figure 2. Some representations of the macrofauna populations in the compared systems: left: Stepped Cascade System; right: Flow Form Cascade System. Species are approximately located in their preferred part of the ecosystem of the pond; their dimensions indicate relative abundance of the species in the compared systems [allows comparison within each species only]. On the far left and far right, those species are shown, that occurred with most individuals either in the Stepped Cascade System (left) or in the Flow Form Cascade System (right) in 1984.

Figure 3. An idea is given of the difference between a more vegetative growth from the Stepped Cascade ponds (2 upper rows) and a more generative growth from the Flow Form Cascade ponds (3 lower rows). Plants are shown in different stages of development, going from left (June) to right (August). (See below)

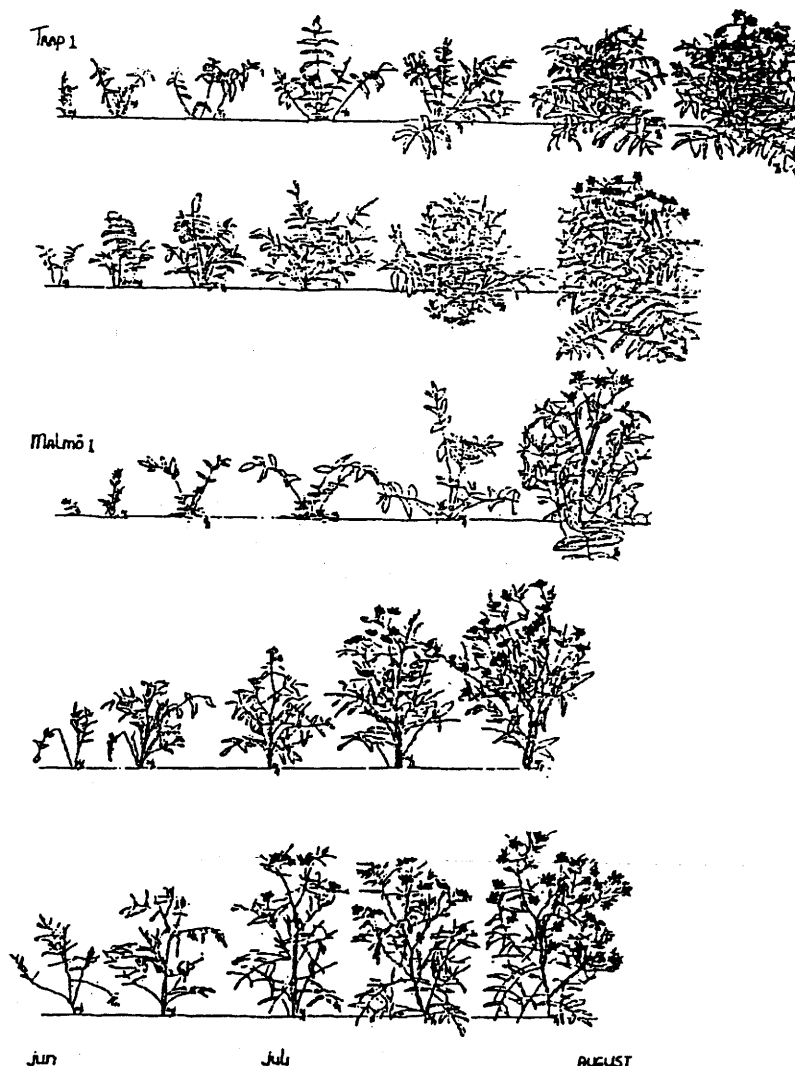


Figure 3.